



QEMU

Part—4

This is the last article of this series on QEMU. In the previous article, we worked on bare-metal programming, and discussed the need for a boot-loader. Most GNU/Linux distros use GRUB as their boot-loader (earlier, LILO was the choice). In this article, we will test the famous U-Boot (Universal Boot-Loader).

In embedded systems, especially in mobile devices, ARM processor-based devices are leading the market. For ARM, U-Boot is the best choice for a boot-loader. The good thing about it is that we can use it for different architectures like PPC, MIPS, x86, etc. So let's get started.

Download and compile U-Boot

U-Boot is released under a GNU licence. Download it from <ftp://ftp.denx.de/pub/u-boot/>, which has every version of U-Boot available. For this article, I got version 1.2.0 (*u-boot-1.2.0.tar.bz2*). Extract the downloaded tar ball and enter the source code directory:

```
# tar -jxvf u-boot-1.2.0.tar.bz2
# cd u-boot-1.2.0
```

To begin, we must configure U-Boot for a particular board. We will use the same ARM Versatile Platform Baseboard (*versatilepb*) we used in the previous article, so let's run:

```
# make versatilepb_config arch=ARM CROSS_COMPILE=arm-none-eabi-
Configuring for versatile board...
Variant: PB926EJ-S
```

After configuration is done, compile the source code:

```
# make all arch=ARM CROSS_COMPILE=arm-none-eabi-
for dir in tools examples post post/cpu; do make -C $dir _depend; done
make[1]: Entering directory `/root/qemu/u-boot-1.2.0/tools'
ln -s ../common/environment.c environment.c
.
.
G++_lite/bin/.../lib/gcc/arm-none-eabi/4.4.1 -lgcc \
```

```
-Map u-boot.map -o u-boot
arm-none-eabi-objcopy --gap-fill=0xff -O srec u-boot u-boot.srec
arm-none-eabi-objcopy --gap-fill=0xff -O binary u-boot u-boot.bin
```

Find the size of the compiled U-Boot binary file (around 72 KB in my experience) with `ls -lh u-boot*`—we will use it later in this article. I assume that you have set up QEMU, networking and the ARM tool chain, as explained in previous articles in this series. If not, then I suggest you read the last three articles.

Boot U-Boot in QEMU

Now we can boot the U-Boot binary in QEMU, which is simple. Instead of specifying the Linux kernel as the file to boot in QEMU, use the U-Boot binary:

```
# qemu-system-arm -M versatilepb -nographic -kernel u-boot.bin
```

Run some commands in U-Boot, to check if it is working:

```
Versatile # printenv
bootargs=root=/dev/nfs mem=128M ip=dhcp netdev=25,0,0xf1010000,0
xf1010010,eth0
bootdelay=2
baudrate=38400
bootfile="tftpboot/uImage"
stdin=serial
stdout=serial
stderr=serial
verify=n
Environment size: 184/65532 bytes
```

The next step is to boot a small program from U-Boot. In the previous article, we wrote a small bare-metal